

HORIZON-WIDERA-2023-ERA-01-04: Development of new approaches to the macroeconomic modelling of research and innovation

Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 1.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 1.00 million.
<i>Type of Action</i>	Coordination and Support Actions

Expected Outcome: The successful proposal will improve the integration of R&I within macro-economic modelling, while at the same time bringing fresh light on the role of R&I to tackle current societal challenges, with a particular focus on the European context. Project results are expected to contribute to the following expected outcomes:

- Support research at the knowledge frontier of macroeconomic models;

- Contribute to the development of innovative and interdisciplinary approaches to modelling of Research and Innovation (R&I) and R&I policy in Europe;
- Better integrate R&I activities in macro-economic models, leading to better measurement of their impact and a better understanding of the channels through which research and innovation lead to impact (direct and indirect) in Europe;
- Investigate new approaches to R&I macroeconomic models that link R&I-related drivers to economic, scientific and societal outcomes;
- Improve macro-economic modelling for the evaluation of R&I policy on different outcome variables, at different levels of policy intervention (EU, national, regional);
- Produce a novel macro-economic model able to simulate the economy, including the R&I dimension, linking R&I to sustainable and fair growth;
- Provide the full description of the models, codes and datasets, to allow easy access to forefront modelling ideas to the European scientific community and broader audience.

Scope: Existing macroeconomic models still fall short in providing sound ex-ante estimates of the various impacts of R&I. Therefore, the European Commission has engaged in a medium- to long-term agenda to improve the treatment of research and innovation (R&I) in macroeconomic models by supporting and financing cutting edge research on macroeconomic modelling able to attract the best researchers to work and develop new models capable of supporting the rising demand for policy evaluation tools in the EU. This agenda started in 2014 following the Commission Communication "Research and innovation as sources of renewed growth". The Council of the European Union called on the Commission to promote further research to build the evidence base for R&I policy making and for budgetary decision making, including by further improving the way R&I is accounted for in macro-economic models.

On the one hand, R&I are acknowledged as key engines for long-term growth and societal prosperity by an extensive body of economic literature. On the other hand, there is still need to develop further evidence about the quality, relevance and impact of R&I investments and the role that related policies have in scientific, technological, economic and social development. There is a crucial need for more developed and nuanced tools that would be able to account for the heterogeneity of different economic actors and that would be able to simulate the effects of a wide range of policies along different dimensions relevant for societal wellbeing. In addition, it is important that models are able to distinguish between the innovation process and the adoption / diffusion process.

Against these backdrops, the proposals, building upon the current state of the art existing in the literature, should develop and provide new macro-economic models with R&I, capable to integrate socio-economic challenges.